JAYAMADU GAMMUNE

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Research Interests

My research focuses on developing autonomous robotic systems that interact safely and efficiently with humans in dynamic environments. I work on UAV autonomy, multi-agent state estimation and cooperative localization, learning-based control for robot manipulation, computer vision and deep learning for perception, and vision-language models for task understanding. My interests span mechatronic system design including UAVs, unmanned vehicles, and soft robotics, with emphasis on developing novel robotic structures and mechanisms.

Education

B.Sc. Engineering (Hons.) in Biomedical Engineering — University of Moratuwa *2022 – 2026*
First Class Honours Specialization: Robotics & Automation Pathway GPA: 3.77/4.00 Dean's List: Semesters 4, 6, 7, 8

Relevant Coursework: Advanced Electronic Control Systems, Autonomous Systems, Deep Learning for Computer Vision, Robotics, Electronic Control Systems, Embedded Systems, Data Structures and Algorithms

General Certificate of Education Advanced Level *2020*
Physical Science Stream (English Medium) 3 Distinctions Z-Score: 2.3204

Experience

Intern Research Affiliate *2024 – 2025*
Intelligent Robotics Laboratory, Swinburne University of Technology, Australia
Advisor: Assoc. Prof. Mats Isaksson

- Developed GPU-accelerated computer vision algorithms for autonomous robotic ultrasound imaging, achieving 27% reduction in processing latency for real-time anatomical landmark detection using PyTorch and CUDA.

Visiting Instructor *Sep – Nov 2025*
Dept. of Electronic and Telecommunication Engineering, University of Moratuwa, Sri Lanka

- Instructed laboratory sessions for EN 2091 Laboratory Practice and Projects, covering embedded systems, electronic circuits, and robotics; provided hands-on guidance in hardware integration and system design for undergraduate students.

Visiting Instructor (Teaching Assistant) *Jun – Oct 2026*
Dept. of Electronic and Telecommunication Engineering, University of Moratuwa, Sri Lanka

- Served as Teaching Assistant for EN 3563 Robotics, EN 4594 Autonomous Systems, and EN 2091 Laboratory Practice and Projects by conducting sessions, supporting student learning, and assisting with hardware integration and system design.

Publications & Presentations

- D.J.T. Gammune, M.C. Roshan, and M. Isaksson, "Systematic GPU Acceleration for Autonomous Robotic Ultrasound Imaging via Point Cloud Pipeline Optimization," in *Proc. MERCon 2026 (Moratuwa Engineering Research Conference)*, University of Moratuwa, Sri Lanka, 2026. doi: 10.1109/MERCon71835.2026.11691436.
- D.J.T. Gammune, R.D. Kumarasinghe, C.N. Weerasinghe, W.K.D.D. Wijewickrama, P. Kulasingham, C. Edussooriya, P. Jayasekara, J. Wanigasinghe, and K. Wickremasinghe, "Development of a Child-Friendly Non-Invasive Brain-Computer Interface for Paediatric Locked-in Syndrome," in *Proc. 19th Asia Pacific Congress of Paediatrics (APCP 2026)*, Colombo, Sri Lanka, Jul. 2026. [**Abstract**]

Research Projects

Extending RoboBrain 2.0 with an LLM Planning Layer [GitHub] *2026*
2-member research collaboration
Stack: PyTorch, Transformers, CUDA, bitsandbytes, Flask, React

- Extended the published RoboBrain baseline (CVPR 2025) by integrating a dedicated LLM layer for complex hierarchical task decomposition.
- Developed hybrid architecture decoupling high-level planning from low-level control, boosting accuracy on long-horizon tasks.
- Engineered an 8-bit quantization pipeline reducing memory usage by 50%, enabling real-time deployment on consumer hardware.

GPU-Accelerated Autonomous Robotic Ultrasound System *2024 – 2025*
Intelligent Robotics Laboratory, Swinburne University of Technology (remote, from Sri Lanka)
Stack: ROS, PyTorch, CUDA, OpenCV, Python

- Developed CUDA-accelerated image processing algorithms achieving 27% latency improvement in lumbar spine landmark detection.
- Optimized control architectures integrating OpenCV pipelines for closed-loop medical imaging feedback and safe clinical human-robot interaction.

Field-Oriented Control for Robotic Actuators *Ongoing*
3-member research team
Stack: STM32, Motor Control Theory, Power Electronics, Real-Time Systems

- Implementing FOC algorithms on STM32 microcontrollers for precise torque regulation in BLDC motors.
- Designed custom motor drivers with optimized power electronics, used in the Micromouse platform (see Technical Projects).

Paediatric EEG-Based Brain-Computer Interface for Assistive Robotics *2026*
4-member interdisciplinary team
Stack: Assistive Robotics, Neural Signal Processing, Embedded Systems

- Designed a paediatric BCI system controlling a simple robotic arm via real-time EEG processing.
- Developed custom low-noise PCBs for biopotential amplification and ML/signal processing models for robust, non-invasive intention recognition.

Technical Projects

Survey Drone with Aerial Image Capture and Map Reconstruction *Ongoing*
2-member team, 2 supervisors
Stack: ArduPilot, Pixhawk

- Building a survey drone platform for aerial data collection, including automated image capture over a survey area.
- Developing a map reconstruction pipeline that stitches the captured aerial images into a map of the surveyed area.

Vision-Guided Robotic Pick-and-Place System [GitHub] *2024*
4-member design team
Stack: Altium Designer, SolidWorks, OpenCV, Python, C++

- Integrated mechatronic system combining SolidWorks mechanical design, Altium custom PCB, and OpenCV computer vision for precision object manipulation.
- Implemented real-time visual servoing with cross-hair overlay for accurate hole alignment and autonomous placement control.

Autonomous Maze-Solving Micromouse *2025*
3-member team
Stack: STM32 (CubeIDE), Altium Designer, SolidWorks, FloodFill Algorithm, Gazebo, Webots

- Engineered high-speed autonomous robot using FloodFill algorithm with optimized speed and traversal efficiency for complex maze navigation.
- Designed compact mechanical chassis and custom power distribution PCBs.

Multi-DOF Robotic Arm: Kinematics and Control [GitHub] *2024*
3-member team
Stack: MATLAB Robotics Toolbox, Python, C++

- Implemented forward/inverse kinematics with trajectory planning for multi-DOF manipulator using MATLAB Robotics Toolbox.
- Developed real-time control interface with emphasis on precision positioning and motion stability.

Autonomous Mobile Robot for Competition [GitHub] *2024*
Sri Lankan Robotics Challenge (SLRC), 5-member team
Stack: Arduino, C++, Sensor Integration, Path Planning

- Designed and implemented autonomous mobile robot with integrated embedded control systems for national competition.
- Developed sensor fusion and navigation algorithms for competitive performance in dynamic environments.

Multi-Tasking Autonomous Platform [GitHub] *2023*
4-member team
Stack: ATmega2560, C/C++, PID Control, SolidWorks, Sensor Fusion

- Developed ATmega2560-based mobile robot with custom chassis and mechanical arm featuring PID control for line following and wall avoidance.
- Integrated sensor logic for color detection, sound frequency sensitivity, and ramp navigation.

Computer Vision Human Tracking System [GitHub] *2024*
Stack: Python, OpenCV, Tkinter

- Developed real-time human detection and tracking system with GUI for visualization; applicable to surveillance and mobile robotics.

Technical Competencies

Robotics & Control	ROS, Kinematics and Dynamics, Trajectory Planning, Motion/Model-Based Control, Field-Oriented Control, Gazebo Simulation, MATLAB Robotics Toolbox, Path Planning, Mobile Robotics
Computer Vision	OpenCV, Real-time Object Detection and Tracking, Visual Servoing, GPU-Accelerated Vision, CUDA Programming, Image Processing, Deep Learning for Perception
Machine Learning & AI	PyTorch, Transformers, Vision-Language Models, Deep Learning, Neural Networks, Model Optimization and Quantization, GPU Acceleration
Embedded Systems	STM32, AVR/ARM Microcontrollers, RTOS, Firmware Development (C/C++), Motor Control, Sensor Integration, Embedded Signal Processing
Electronics Design	Altium Designer (PCB Layout), Analog Circuit Design, Power Electronics, Motor Driver Design, Biopotential Amplifiers, SPICE Simulation, EMI/EMC
Programming	Python, C/C++, CUDA, JavaScript/React, MATLAB, SystemVerilog
Tools	Git, Linux/Ubuntu, SolidWorks (CAD), Proteus, Multisim

Professional Development

Building Transformer-Based NLP Applications	NVIDIA	2026
Introduction to Deep Learning for Computer Vision	MathWorks	2025
Deep Learning for Object Detection	MathWorks	2025
Advanced Deep Learning Techniques for Computer Vision	MathWorks	2025
Teaching Old Motors New Tricks: Field-Oriented Control	Texas Instruments	2025
Supervised Machine Learning: Regression and Classification	Stanford / DeepLearning.AI	2025
High-Performance C++ Development	LSEG	2024

Service & Positions

Web Manager	IEEE Sri Lanka Section	Ongoing
Web Manager	IEEE EMBS Sri Lanka Chapter	2025 – 2026
Web Master	IEEE EMBS Student Chapter, Univ. of Moratuwa	2024 – 2025
Assistant Web Master	IEEE EMBS Student Chapter, Univ. of Moratuwa	2024 – 2025
Panelist (Panel Discussion)	AXON – IEEE EMBS Sri Lanka Chapter	2024
Design Team	Electronic Club, ENTC	2022 – 2024
Demonstrator	EXMO, University of Moratuwa	2023
Educational Outreach Volunteer	"Soyuru Sathkara" & "Sasnaka Sansada"	2022 – 2024
Under-15 Cricket & Scrabble	Prince of Wales College, Moratuwa	2014 – 2015

Additional Competencies

Languages: Sinhala (Native), English (IELTS Academic: 7.5)
Communication: Technical presentation, scientific writing, research documentation
Collaboration: International research collaboration, cross-functional teamwork, mentoring

References

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